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Blind Image Deconvolution: When Patch-wise Minimal Pixels Prior Meets Fractional-Order Method

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Abstract
Blind image deconvolution is a challenging issue in image processing. In blind image deconvolution, the typical approach involves iteratively estimating both the blur kernel and latent image until convergence to the blur kernel of the observed image is achieved. Recently, several approaches have been attempted to develop a sophisticated regularization to obtain the clean image. However, existing methods often struggle to effectively handle ringing artifacts and local blur. To overcome these limitations, we introduce a fractional-order variational model. This model alleviates the ringing artifacts through the selection of an optimal derivative. Subsequently, to refine the latent image further, we leverage the local prior, namely patch-wise minimal pixels (PMP) prior. Since the PMP prior of clean images blocks is much sparser than that of blurred ones, it is capable of discriminating between clean and blurred image blocks. We illustrate the effective integration of the fractional-order operations and the PMP prior within our proposed approach. Moreover, the convergence of our algorithm has been proved as the values of the objective function monotonically decrease. Extensive experiments on different datasets demonstrate the superiority of the proposed method compared with other methods in terms of reconstruction quality for blind deconvolution.

Keywords Image deconvolution · Blind deconvolution · Fractional order · Ringing artifacts

1 Introduction

Blind image deblurring is a foundation problem in image processing. It is widely used in computer vision, pattern recognition, robotic grasping, and autonomous driving. In this study, we concentrate on the blind deconvolution problem within the realm of image deblurring, with a particular emphasis on uniform blur. We believe that addressing the issue of uniform blur remains an important and significant topic with practical implications and research value for several reasons. Firstly, uniform blur is one of the most common types of blur encountered in practical applications, especially in images captured under conditions due to camera shake. Secondly, while most current deblurring methods have made progress in dealing with spatially variant blur, there is still room for improvement in the quality of deblurring for images with uniform blur. Furthermore, uniform blur is often considered a subset of the deblurring problem due to its relatively simple blur kernel, making it valuable to develop more efficient algorithms for this specific issue. Lastly, deblurring of images with uniform blur can be seen as a cornerstone for more complex blur problems, providing a foundation for tackling more challenging scenarios.

Considering a spatially invariant blur kernel k and a latent sharp image I, the corresponding blurred image B can be defined as:

$$B = k \otimes I + \varepsilon, \tag{1}$$

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where $\otimes$ denotes the convolution operator and ε is the additive white Gaussian noise. We assume that $B, I \in \mathbb{R}^{m\times n}$, $k \in \mathbb{R}^{l\times l}$ and each element of the matrix is between 0 and 1. We assume that B is uniform; hence, the blur kernel matrix with dimensions $l \times l$ is presumed to be considerably smaller relative to the image matrix with dimensions $m \times n$, and $\sum_{(i,j)\in supp(k)} k_{i,j} = 1$, where $supp(k)$ denotes the support of blur kernel. In general, the blurry image is recovered by solving the following problems:

$$\arg\min_{I} \ \|k \otimes I - B\|_2^2 + \mu\, R(I) \tag{2}$$

and

$$\arg\min_{k} \ \|k \otimes I - B\|_2^2 + \gamma\, D(k), \tag{3}$$

where $R(I) : \mathbb{R}^{m\times n} \to \mathbb{R}$ and $D(k) : \mathbb{R}^{l\times l} \to \mathbb{R}$ are regularizations or priors on the latent image and blur kernel and μ and γ are positive weight parameters. Traditional methods address the blind deconvolution problem by appropriately changing these regularization terms.

Accordingly, several attempts have been made to construct the expressions for $R(I)$ and $D(k)$ in (2) and (3), respectively, to improve the accuracy of blur kernel estimation [1–4]. When employing a known blur kernel, (1) evolves into a non-blind deconvolution problem. Subsequently, the application of non-blind deblurring techniques [5] allows attaining superior deblurring outcomes. In essence, refining the accuracy of the blur kernel imparts a high specificity level to the problem, facilitating the acquisition of enhanced deblurred images through non-blind methodologies. In this context, our objective is to enhance the precision of the blur kernel obtained through the blind deconvolution process described by (2) and (3).

Most methods are premised on the integer-order gradient (∇^α, $\alpha = 1$) of image as $R(I)$ enables computational efficiency. However, it may be affected by local information and lead to the so-called 'ringing artifacts' [6]. Ringing artifacts typically manifest as false edges around abrupt transitions in an image, creating a visual effect that resembles ripples or ghostly bands. Therefore, the focus of some studies has shifted toward fractional-order (∇^α, $\alpha \in (1, 2)$) computation. For example, Li et al. [7] employed an adaptive fractional-order differential function to enhance medical images, where the parameter α can take any value between one and two for ∇^α, offering global properties of an image in contrast with the local properties delivered by the integer-order derivative (a detailed explanation of fractional-order derivative in Sect. 3). As illustrated in Fig. 1, the fractional-order derivative of pixel $I_{i,j}$ is determined by more extensive neighboring pixels (in comparison to the integer order), enabling it to capture more image information. Furthermore, it can mitigate the likelihood of generating the undesired ringing artifacts inherent in the algorithm. According to this feature, we replace the normally used ∇I with $\nabla^\alpha I$ as $R(I)$ for the blind deconvolution problem.

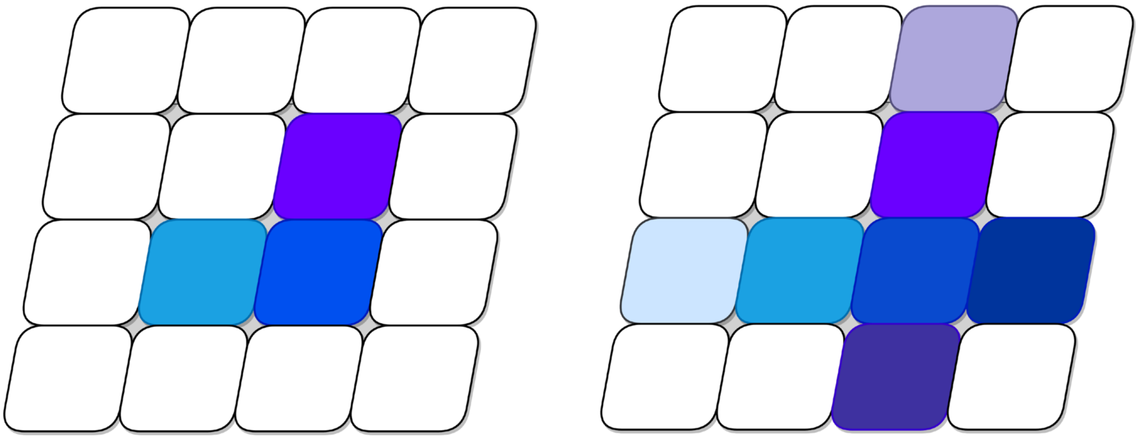
This paper makes the following contributions: Firstly, we introduce a novel blind image deconvolution model that utilizes a fractional-order gradient operator and the PMP prior. Secondly, we present a blind deconvolution algorithm designed to address the proposed model, with demonstrated convergence and a consistent decrease in the values of the objective function. Thirdly, we illustrate the superiority of our approach on widely used natural images datasets by Köhler et al. [8], Levin et al. [9] and Lai et al. [10]. Regarding real-world images, experimental results demonstrate that our algorithm can compete with the state-of-the-art methods.

The paper is organized as follows. In Sect. 2, we present the background material for this work. In Sect. 3, we present the preliminary information relevant to this paper. The new algorithm is developed in detail in Sect. 4. In Sect. 5, we offer a proof of convergence with certain variables for the iteration generated by our algorithm, with monotonically decreasing objective function values. In Sect. 6, we showcase various experimental results that demonstrate the effectiveness of the proposed method. Finally, in Sect. 7, we conclude this paper.

2 Related Work

Over the past few years, significant strides have been made in the field of image blind deconvolution [8, 11]. This significant progress is primarily because of the incorporation of statistical priors derived from natural images and the emphasis identifying salient edges to accurately estimate the blur kernel [4, 12, 13]. For instance, Cho and Lee [1] leveraged edge information of the latent image to suppress artifacts and reduce noise in the intermediate process, thereby improving the accuracy of blur kernel estimation. Another approach, employing Gaussian prior for latent images and supplemented by an additional edge selection procedure, was implemented to enhance the precision of blur kernel estimation [4]. However, these methods were susceptible to certain limitations, such as reliance on heuristic filters and assumptions of well-defined edges in the latent images, which may not always hold in the edge selection step.

To overcome these constraints and improve the recovery of sharp edges during blur kernel estimation simultaneously, many researchers have developed exemplar-based methods [14, 15]. These methods can extract meaningful information from both blurry and example images to enhance edge recovery quality. Further, Fergus et al. [16] proposed a deconvolution algorithm based on the gradient distribution model. The work of Levin et al. [17] highlighted the advantages of the variational Bayesian inference method [16] in avoiding



$$\nabla_h I_{3,3} = I_{3,3} - I_{2,3}$$

$$\nabla_v I_{3,3} = I_{3,3} - I_{3,2}$$

(a) Integer-order derivative

$$\nabla_h^{\alpha} I_{3,3} = c_1 I_{1,3} + c_2 I_{2,3} + c_3 I_{3,3} + c_4 I_{4,3}$$

$$\nabla_v^{\alpha} I_{3,3} = c_1 I_{3,1} + c_2 I_{3,2} + c_3 I_{3,3} + c_4 I_{3,4}$$

(b) Fractional-order derivative

Fig. 1 Difference between the integer-order derivative and the fractional-order derivative in the horizontal and vertical directions. (a) and (b) represent the pixel points used for the integer order and the fractional order at point $I_{3,3}$, respectively. Specifically, the calculation of fractional order involves more pixels than the integer order

trivial solutions, in contrast with the limitations encountered with naive maximum a posteriori (MAP) methods. A remarkable blind image deconvolution technique was proposed by Xu et al. [18], who utilized a piecewise function to approximate the $\mathcal{L}_0$-norm and robustly model the distribution of image gradients. This method demonstrated notable performance in image restoration. A dark channel prior (DCP) was introduced which combines intensity information and is designed specifically for various image restoration [3]. Furthermore, Wen et al. [19] proposed a novel regularization approach focusing on the PMP prior to the latent sharp image, leading to an outstanding performance in image restoration. Shao [20] presents a novel regularization approach using redescending potential functions, effectively bridging the gap between blind and non-blind deblurring and boosting performance. However, these methods are susceptible to interference from local pixels information in images, which implies their models tend to produce some ringing effects and local blur.

To address these issues, Yu et al. [21] and Liu et al. [22] proposed improved methods based on the DCP and the PMP, respectively. These methods incorporate Grunwald–Letnikov (G–L) fractional order into the image priors as a regularization for iterative deconvolution. Since the priors focus on representing local minima within the image, the fractional-order is designed to capture the global gradient characteristics. Therefore, integrating these two approaches may lead to mutual interference. On the one hand, it may weaken the effectiveness of the prior information, and on the other hand, it could limit the global expressive power of the fractional-order. Additionally, these methods have not fully considered the convergence of the algorithms, which could affect the stability and reliability of the deconvolution performance.

In recent years, robust deep learning strategies have led to the widespread adoption of data-driven methods, as they have been instrumental in achieving significant advancements [23, 24]. Mao et al. [25] introduced an idempotent deep network and an efficient deblurring model for superior blind image deblurring with reduced parameters and faster speeds. Kaufman et al. [26] proposed an analysis-synthesis network architecture for estimating image blur kernels and utilizing them to restore clear images. Carbajal et al. [27] further introduced a novel approach for image deblurring that estimates pixel-wise motional blur kernels using a kernel prediction network and achieves superior deblurring results through a jointly trained deep deblurring network. Despite their effectiveness across various scenarios, it is crucial to note that learning-based methods often heavily rely on training data, which limits the effectiveness of blind deblurring.

3 Preliminaries

In this section, we present two fundamental concepts. Firstly, we elaborate on the PMP prior which serves as a robust metric for discriminating between clean and blurred images. Secondly, we delve into the fractional-order gradient operator which plays a pivotal role in the model.

3.1 Patch-Wise Minimal Pixels

The PMP prior represents the simplified sparsity prior of local minimal pixels; the PMP prior of the image is defined in [19] as follows.

Definition 1 (PMP prior) Given an image $I \in \mathbb{R}^{m\times n\times 3}$, the PMP prior is:

$$\mathcal{P}(I)(i) = \min_{(x,y)\in\psi_i}\left(\min_{z\in\{r,g,b\}} I(x,y,z)\right), \tag{4}$$

where $i = 1, 2, \ldots, P$, ψ_i is one of the P non-overlapping blocks into which I is partitioned, and z denotes the color channel. When I is a grayscale image, $\min_{z\in\{r,g,b\}} I(x,y,z) = I(x,y)$.

Given an image $I \in \mathbb{R}^{m\times n}$, the PMP prior function is represented as $\mathcal{P}(I) : \mathbb{R}^{m\times n} \to \mathbb{R}^P$. We introduce its inverse operation for subsequent utilization. Its inverse is denoted by its transpose $\mathcal{P}^T(\vec{v}) : \mathbb{R}^P \to \mathbb{R}^{m\times n}$ for any $\vec{v} \in \mathbb{R}^P$. Then we have

$$\mathcal{P}^T(\mathcal{P}(I)) = I \circ M, \tag{5}$$

where $\circ$ represents dot multiplication and $M \in \mathbb{R}^{m\times n}$ represents the binary mask corresponding to the subset of I defined by the PMP. Specifically, $M(i,j) = 1$ if $I(i,j)$ denotes a minimal pixel of a patch, and $M(i,j) = 0$ otherwise.

From an intuitive standpoint, the blurring process exerts a smoothing influence on image pixel intensities. Consequently, the patch-wise minimal pixels tend to increase after applying the blur operation. The PMP prior contains patch-wise minimal pixels of the images; therefore, the PMP of clean images is much sparser than those of blurred images.

This feature is not only observed in natural images but also in other types of images such as text and low-light images (please refer to Sect. 6). Hence, the sparsity property of the PMP prior serves as an effective method for distinguishing clean images from blurred ones.

3.2 Fractional-Order Gradient

The fractional-order derivative extends the integer-order derivative to allow arbitrary orders [28–31]. Three main definitions are used to define the fractional-order derivative: the Riemann–Liouville (R–L), the G–L, and the Caputo definitions. Each provides a different approach to perform calculations involving fractional order. Among these, the R–L and the Caputo fractional-order derivatives are both refinements built upon the foundation laid by the G–L fractional-order derivative. Notably, the R–L fractional-order derivative has been subjected to rigorous scrutiny and validation within the domain of image processing [28, 32]. In this work, we primarily utilize the R–L definition for our calculations and analyses.

Definition 2 (R–L fractional-order)[33] For a function $f(t)$, the left and right R–L fractional-order derivatives are defined as:

$$D^{\alpha}_{[a,t]} f(t) = \frac{1}{\Gamma(v-\alpha)}\frac{d^v}{dt^v}\left(\int_a^t \frac{f(\tau)}{(t-\tau)^{\alpha-v+1}}d\tau\right) \tag{6}$$

and

$$D^{\alpha}_{[t,b]} f(t) = \frac{(-1)^v}{\Gamma(v-\alpha)}\frac{d^v}{dt^v}\left(\int_t^b \frac{f(\tau)}{(\tau-t)^{\alpha-v+1}}d\tau\right), \tag{7}$$

where $v-1 < \alpha \le v$, v is a positive integer, t belongs to the interval between a and b, $\Gamma(\cdot)$ represents the gamma function.

Building upon the above-mentioned definitions, we can have the Riesz–R–L (central) fractional-order derivative as follows.

Definition 3 (Riesz–R–L fractional-order)[34] Let $v-1 < \alpha \le v$ and v be a positive integer, for a function $f(t)$, the Riesz–R–L (central) fractional-order derivative is defined as:

$$D^{\alpha}_{[a,b]} f(t) = \frac{1}{2}\left[D^{\alpha}_{[a,t]} f(t) + D^{\alpha}_{[t,b]} f(t)\right]. \tag{8}$$

As our focus is on images, we compute the fractional-order derivative for discrete functions. According the references [35, 36], we consider an image represented by a matrix $I \in \mathbb{R}^{m\times n}$ under the Dirichlet boundary condition. For $\alpha \in (1,2)$, ∇^{α} denotes the operator of a fractional-order differential of order α. The discrete fractional-order gradient at the point (i,j) in the horizontal and vertical directions can be expressed as:

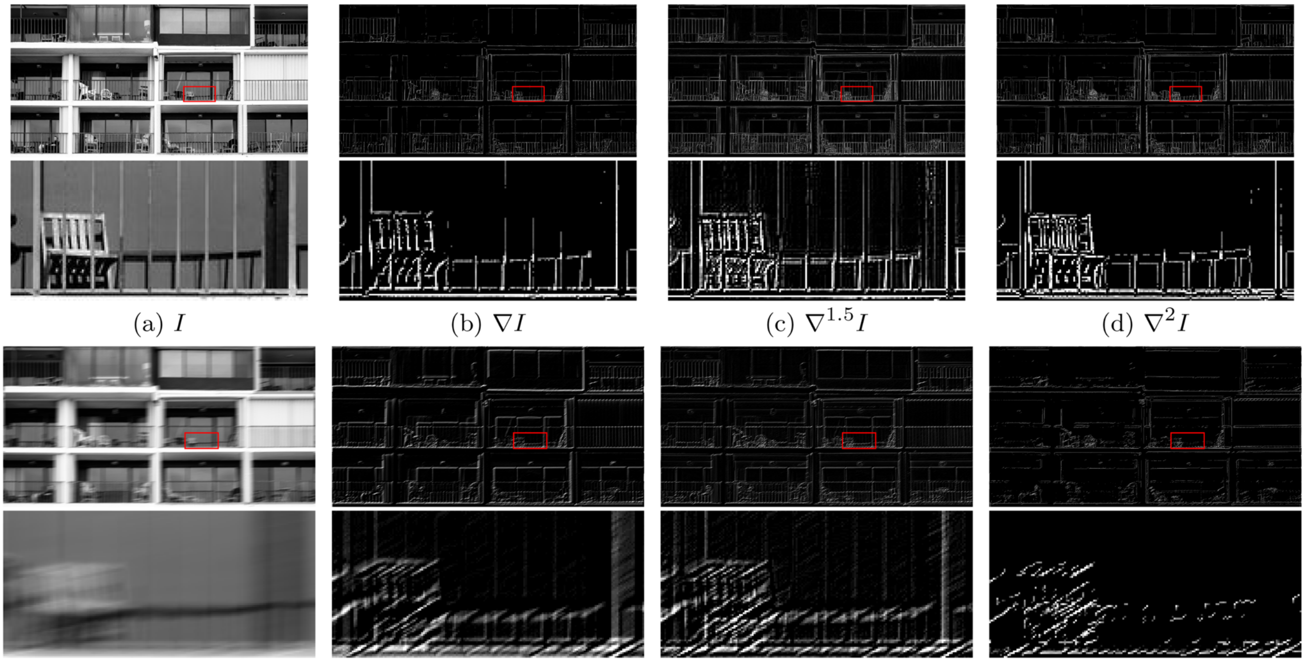
$$\nabla^{\alpha}_h I_{i,j} = \frac{1}{2}\left[\sum_{k=0}^{i} C^{\alpha}_k I_{i-k+1,j} + \sum_{k=0}^{m-i+1} C^{\alpha}_k I_{i+k-1,j}\right], \tag{9}$$

$$\nabla^{\alpha}_v I_{i,j} = \frac{1}{2}\left[\sum_{k=0}^{j} C^{\alpha}_k I_{i,j-k+1} + \sum_{k=0}^{n-j+1} C^{\alpha}_k I_{i,j+k-1}\right], \tag{10}$$

where i ranges from 1 to m and j ranges from 1 to n. The value of parameter C^{α}_k can be determined using the following recursive formulas:

$$C^{\alpha}_0 = 1,\, C^{\alpha}_k = (1 - \frac{1+\alpha}{k})C^{\alpha}_{k-1}, \tag{11}$$

where $k = 1, 2, \ldots$. By setting $C = C^{\alpha}_0 + C^{\alpha}_2$, we can now define the discrete fractional-order gradient of the image I.



(a) I (b) ∇I (c) $\nabla^{1.5} I$ (d) $\nabla^2 I$

(e) B (f) ∇B (g) $\nabla^{1.5} B$ (h) $\nabla^2 B$

Fig. 2 Different gradients of the clean image I and blurred image B. From left to right, we illustrate order 0, order 1, order 1.5, and order 2 of the image, respectively

Definition 4 (Discrete fractional-order) Let $I, I_h, I_v \in \mathbb{R}^{m\times n}$ represent images under the Dirichlet boundary condition. For $\alpha \in (1, 2)$, the discrete fractional-order gradient of I is:

$$\nabla^\alpha I = \begin{pmatrix} M_\alpha I \\ I N_\alpha^T \end{pmatrix}, (\nabla^\alpha)^T \begin{pmatrix} I_h \\ I_v \end{pmatrix} = M_\alpha^T I_h + I_v N_\alpha, \tag{12}$$

where

$$M_\alpha = \frac{1}{2}\begin{bmatrix} 2C_1^\alpha & C & C_3^\alpha & \cdots & C_m^\alpha \\ C & 2C_1^\alpha & \ddots & \ddots & \vdots \\ C_3^\alpha & \ddots & \ddots & \ddots & C_3^\alpha \\ \vdots & \ddots & \ddots & 2C_1^\alpha & C \\ C_m^\alpha & \cdots & C_3^\alpha & C & 2C_1^\alpha \end{bmatrix}, \tag{13}$$

$$N_\alpha = \frac{1}{2}\begin{bmatrix} 2C_1^\alpha & C & C_3^\alpha & \cdots & C_n^\alpha \\ C & 2C_1^\alpha & \ddots & \ddots & \vdots \\ C_3^\alpha & \ddots & \ddots & \ddots & C_3^\alpha \\ \vdots & \ddots & \ddots & 2C_1^\alpha & C \\ C_n^\alpha & \cdots & C_3^\alpha & C & 2C_1^\alpha \end{bmatrix}. \tag{14}$$

For an image I, the first-order formulas in the horizontal and vertical directions at point (i, j) are as follows:

$$\nabla_h I_{i,j} = I_{i+1,j} - I_{i,j} \quad \text{and} \quad \nabla_v I_{i,j} = I_{i,j+1} - I_{i,j}, \tag{15}$$

whereas the second-order formulas at that point are:

$$\begin{aligned} \nabla_{hh}^2 I_{i,j} &= I_{i+2,j} - 2I_{i+1,j} + I_{i,j}, \\ \nabla_{vv}^2 I_{i,j} &= I_{i,j+2} - 2I_{i,j+1} + I_{i,j}, \\ \nabla_{hv}^2 I_{i,j} = \nabla_{vh}^2 I_{i,j} &= I_{i+1,j+1} - I_{i+1,j} - I_{i,j+1} + I_{i,j}. \end{aligned} \tag{16}$$

Comparison between (9)-(10) and (15)-(16), we discern that $\nabla^\alpha I$ yields a broader range of global information at point (i, j), spanning both the horizontal and vertical axes, thereby surpassing the information acquisition potential of commonly used integer orders such as ∇I and $\nabla^2 I$.

Refer to Fig. 2 for a visual representation of both the integer-order and the fractional-order gradients of the clean image I and blurred image B. It is apparent that the fractional-order gradients convey more image information and exhibit finer details as contrasted with the first-order gradients in images. Moreover, fractional-order gradients offer smoother edges and better handling of details than second-order counterparts. Consequently, it is plausible to surmise that applying fractional order could potentially outperform integral-order approaches in blind deconvolution problems.

4 Model and Optimization

In this section, we present the proposed model and the detailed algorithm to solve the proposed model. In the first part, we focus on the formulation of the MAP blind deconvolution problem. Then, we employ the half-quadratic splitting algorithm to estimate both the image and the blur kernel.

4.1 Numerical Scheme of the Blind Deconvolution Model

We begin by recalling the blind image deconvolution formulation, the MAP formulation, represented by:

$$\min_{I,k} \|k \otimes I - B\|_2^2 + \gamma\, D(k) + \mu\, R(I), \tag{17}$$

where I, B, and k are matrices with elements between 0 and 1. The first term is the fidelity term, which ensures that the convolution of the estimated blur kernel k and latent image I approximates the observed blurred image B. In order to tackle the ill-posed nature of the problem, we introduce the image prior $R(I) : \mathbb{R}^{m\times n} \to \mathbb{R}$ and the blur kernel prior $D(k) : \mathbb{R}^{l\times l} \to \mathbb{R}$ into the formulation [37–40]. Neglecting these priors would render the problem ill-posed, as it would allow for numerous combinations of k and I that could produce the same blurred image B. Therefore, to enhance the accuracy of the blur kernel estimation and improve image clarity, we add the second and third terms to the equation to regularize the solution for the blur kernel and latent image, respectively.

Generally speaking, in pursuit of a more accurate blur kernel estimation, the second term is selected as $\mathcal{L}_2$-norm penalty of k. Simultaneously, considering the inherent sparsity in the gradients of natural images, the third term is selected as the $\mathcal{L}_0$-norm penalty of ∇I. Then the formula (17) can be rewritten as:

$$\min_{I,k} \|k \otimes I - B\|_2^2 + \gamma\, \|k\|_2^2 + \mu\, \|\nabla I\|_0 . \tag{18}$$

In this paper, to achieve better blind deconvolution results, we apply the $\mathcal{L}_0$-norm penalty of the PMP prior and $\nabla^\alpha I$ as $R(I)$. Based on this, we present a straightforward and effective blind image deconvolution model which is:

$$\min_{I} \|k \otimes I - B\|_2^2 + \gamma\, \|k\|_2^2 + \mu \left\|\nabla^\alpha I\right\|_0 + \|\mathcal{P}(I)\|_0 , \tag{19}$$

where γ and μ are positive weight parameters, α is fractional-order parameter between 1 and 2. The third term utilizes the $\mathcal{L}_0$-norm penalty of $\nabla^\alpha I$ to preserve more image details and avoid the image artifacts, and the last term is imposed to measure the sparsity property of the PMP prior in clean images.

4.2 Algorithm

It is difficult to compute (19) due to the formula involving $\mathcal{L}_0$-norm and $\mathcal{P}(\cdot)$ as the nonlinear operator. To address this, we use the half- quadratic splitting algorithm, introducing auxiliary variables G and w to solve the $\mathcal{L}_0$ terms. By rephrasing the objective function, we ensure $G \approx \nabla^\alpha I$ and $w \approx I$. Next, we will consider minimizing the following model:

$$\begin{aligned} L(G, w, I, k) = {} & \|k \otimes w - B\|_2^2 + \lambda_1 \left\|\nabla^\alpha I - G\right\|_2^2 \\ & + \|\mathcal{P}(I)\|_0 + \mu\, \|G\|_0 + \gamma\, \|k\|_2^2 + \lambda_2\, \|I - w\|_2^2 , \end{aligned} \tag{20}$$

where penalty parameters $\lambda_1, \lambda_2 > 0$ are chosen to enforce $\|\nabla^\alpha I - G\|_2^2 \approx 0$ and $\|I - w\|_2^2 \approx 0$.

The method for (20) entails minimizing L by iteratively adjusting each of the variables, i.e., G, w, I, and k, while holding the others constant at their latest values:

$$G^{t+1} = \arg\min_{G} L(G, w^t, I^t, k^t), \tag{21a}$$

$$w^{t+1} = \arg\min_{w} L(G^{t+1}, w, I^t, k^t), \tag{21b}$$

$$I^{t+1} = \arg\min_{I} L(G^{t+1}, w^{t+1}, I, k^t), \tag{21c}$$

$$k^{t+1} = \arg\min_{k} L(G^{t+1}, w^{t+1}, I^{t+1}, k). \tag{21d}$$

In the domains of image processing, Toeplitz matrices serve to represent linear systems associated with convolution operations. Since motional blur is generated by camera translation which is a linear translation invariant [18], we utilize Toeplitz matrices K and W of k and w, respectively .

In the following, we tackle (21a–d) step by step, addressing each equation separately. Based on the initial estimation of the image I^t and the fact that (21a) is an element-wise minimization problem, we iteratively optimize the G^{t+1} using the following equation:

$$G^{t+1} = \begin{cases} 0, & (\nabla^\alpha I^t)^2 < \mu/\lambda_1, \\ \nabla^\alpha I^t, & \text{otherwise.} \end{cases} \tag{22}$$

The proof that it holds is provided in Appendix A. According to [19, 41], the proximal minimization problem of (21b) can be explicitly obtained as follows:

$$w^{t+1} = \mathcal{F}^{-1}\left(\frac{\lambda_2 \mathcal{F}\left(I^t\right) + \overline{\mathcal{F}\left(K^t\right)} \circ \mathcal{F}\left(B\right)}{\lambda_2 E + \overline{\mathcal{F}\left(K^t\right)} \circ \mathcal{F}\left(K^t\right)}\right), \tag{23}$$

where E is the identity matrix and $\mathcal{F}(\cdot)$ and $\overline{\mathcal{F}(\cdot)}$ denote the fast Fourier transform and its complex conjugate, respectively. The operation $\mathcal{F}^{-1}(\cdot)$ signifies the inverse fast Fourier transform.

Before considering (21c), we first address the following objective:

$$\tilde{I}^{t+1} = \arg\min_{I} \lambda_1 \left\| \nabla^{\alpha} I - G^{t+1} \right\|_2^2 + \lambda_2 \left\| I - w^{t+1} \right\|_2^2 . \tag{24}$$

We can obtain the solution to this least squares problem by solving the linear equation:

$$(\lambda_1 (\nabla^{\alpha})^T \nabla^{\alpha} + \lambda_2 E) I = \lambda_1 (\nabla^{\alpha})^T G^{t+1} + \lambda_2 w^{t+1}. \tag{25}$$

From Definition 4, it follows that $(\nabla^{\alpha})^T \nabla^{\alpha} I = M_{\alpha}^T M_{\alpha} I + I N_{\alpha}^T N_{\alpha}$. Then by simple operations, we transform the formulas above into the following forms:

$$\mathcal{A} I + I \mathcal{B} = \mathcal{C}, \tag{26}$$

where $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are defined as:

$$\begin{cases} \mathcal{A} = \lambda_1 M_{\alpha}^T M_{\alpha} + \lambda_2 E, \\ \mathcal{B} = \lambda_1 N_{\alpha}^T N_{\alpha}, \\ \mathcal{C} = \lambda_1 (\nabla^{\alpha})^T G^{t+1} + \lambda_2 w^{t+1}. \end{cases} \tag{27}$$

The updating formulas of I can be given by:

$$\tilde{I}^{t+1} = \text{Sylvester}(\mathcal{A}, \mathcal{B}, \mathcal{C}). \tag{28}$$

The term $Sylvester(\cdot)$ refers to the operation aimed at solving the Sylvester equation:

$$\mathcal{A} X + X \mathcal{B} = \mathcal{C}, \tag{29}$$

where $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ are given matrices; X is the unknown matrix. The Sylvester equation can find an accurate solution X that satisfies (29).

Since there exists a sparsity regularization on the PMP prior of I in (21c), we can take a shrinkage [19, 36] step in the iteration. Firstly, we define the PMP prior subset of this image, represented as $I_s^{t+1} := \mathcal{P}(\tilde{I}^{t+1})$, and set a threshold parameter $\lambda > 0$. Then, for each $i \in 1, \ldots, P$, the PMP prior is thresholded as:

$$\tilde{I}_s^{t+1}(i) = \begin{cases} 0, & |I_s^{t+1}(i)| < \lambda, \\ I_s^{t+1}(i), & \text{otherwise}. \end{cases} \tag{30}$$

Algorithm 1 Blind deconvolution algorithm

Define blurred image B
Initialize k with results from the coarser level.
for $i \le max\_iter$ **do**
  while $\lambda_1 < \lambda_{max}$ **do**
    for $t = 0 : J - 1$ **do**
      Solve for G using (22).
      Solve for w using (23).
      Solve for I using (32).
    end for
    $\lambda_1 = \eta \lambda_1$.
  end while
  Solve for k using (34).
end for
return Blur kernel k.

Next, we define ψ^{t+1} as the index set of the PMP prior in $\tilde{I}^{t+1}$. We then create a mask that corresponds to the PMP prior subset, which can be expressed as:

$$M_{i,j}^{t+1} = \begin{cases} 1, & (i, j) \in \psi^{t+1}, \\ 0, & \text{otherwise}. \end{cases} \tag{31}$$

Afterward, we update I using the following formula:

$$I^{t+1} = \tilde{I}^{t+1} \circ (1 - M^{t+1}) + \mathcal{P}^T (\tilde{I}_s^{t+1}), \tag{32}$$

where $\mathcal{P}^T$ is the inverse operation of $\mathcal{P}$ defined in Sec. 3.

As indicated by prior research [3, 42], leveraging the gradient information from the latent image is posited to yield a blur kernel that is more precise and exhibits enhanced stability. Given w^{t+1}, the blur kernel estimation is calculated using the following equation:

$$\min_{k} \left\| \nabla W^{t+1} k - \nabla B \right\|_2^2 + \gamma \left\| k \right\|_2^2 . \tag{33}$$

The quadratic form of (33) allows us to handle it through the fast Fourier transform, obtaining an accurate solution from (34).

$$k^{t+1} = \mathcal{F}^{-1} \left(\frac{\overline{\mathcal{F}\left(\nabla_h W^{t+1}\right)} \circ \mathcal{F}\left(\nabla_h B\right) + \overline{\mathcal{F}\left(\nabla_v W^{t+1}\right)} \circ \mathcal{F}\left(\nabla_v B\right)}{\overline{\mathcal{F}\left(\nabla_h W^{t+1}\right)} \circ \mathcal{F}\left(\nabla_h W^{t+1}\right) + \overline{\mathcal{F}\left(\nabla_v W^{t+1}\right)} \circ \mathcal{F}\left(\nabla_v W^{t+1}\right) + \gamma E} \right). \tag{34}$$

The coarse-to-fine manner is harnessed for deconvolution due to the non-convex nature of the problem. We summarize the iterative process for addressing the blind image deconvolution minimization model (20) in Algorithm 1, which consists of three loops. Extensive numerical investigations demonstrate that convergence within a limited number of iterations is typically observed in the inner loop. To ensure both the efficiency and performance of the algorithm, we have configured parameters as follows: $\eta = 2$ and $J = 3$.

5 Convergence Analysis

Now we show that the sequence of $L(G^t, w^t, I^t, k^t)$ obtained from (21) converges. The initial introduction includes Definition 5 and Theorems 1-2, the proof of which establishes the strong convexity with specific variables.

Definition 5 (Strongly convex function) A function $f(x)$ is considered strongly convex if satisfies:

$$f(y) - f(x) \geq \frac{m}{2}\|y - x\|^2 + \nabla f(x)^T(y - x), \tag{35}$$

where $m > 0$, for all x, y in $dom(f)$.

The theorem presented below forms the cornerstone for constructing our problem formulation, as introduced in Theorem 2.

Theorem 1 *(Conditions equivalent to strong convexity) A function $f(x)$ is considered strongly convex if there exists an $m > 0$ and*

$$\nabla^2 f(x) \geq mE, \tag{36}$$

where E is the identity matrix and all x in $dom(f)$.

Proof See Chapter 5 of [43]. □

Theorem 2 *The above-mentioned objective functions $L(G^{t+1}, w, I^t, k^t)$ and $L(G^{t+1}, w^{t+1}, I^{t+1}, k)$ are strongly convex with respect to w and k.*

Proof The first partial derivative of function (21b) is

$$\nabla_w L(G^{t+1}, w, I^t, k^t) = 2{K^t}^T(K^t w - B) + 2\lambda_2(w - I^t), \tag{37}$$

and the second partial derivative of function (21b) is

$$\nabla_w^2 L(G^{t+1}, w, I^t, k^t) = 2{K^t}^T K^t + 2\lambda_2 E \geq 2\lambda_2 E. \tag{38}$$

Since $\lambda_2 > 0$, and G^{t+1}, I^t, k^t are constants, it follows from Theorem 1 that the function $L(G^{t+1}, w, I^t, k^t)$ is strongly convex with respect to the variable w. Similarly, it can be proved that $L(G^{t+1}, w^{t+1}, I^{t+1}, k)$ is strongly convex with respect to variable k. □

Subsequently, we demonstrate the monotonicity and boundedness of the objective function.

Theorem 3 *If the sequence $\{G^t, w^t, I^t, k^t\}$ is generated by the above formula (21), then there exists a constant $C > 0$ such that*

$$\begin{aligned} &L(G^t, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1}) \\ &\geq C\left(\left\|w^{t+1} - w^t\right\|_2^2 + \left\|k^{t+1} - k^t\right\|_2^2\right). \end{aligned} \tag{39}$$

Proof We can decompose the following equation

$$\begin{aligned} &L(G^t, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1}) \\ &= L(G^t, w^t, I^t, k^t) - L(G^{t+1}, w^t, I^t, k^t) \\ &+ L(G^{t+1}, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^t, k^t) \\ &+ L(G^{t+1}, w^{t+1}, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^t) \\ &+ L(G^{t+1}, w^{t+1}, I^{t+1}, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1}). \end{aligned} \tag{40}$$

And based on (21a), G^{t+1} is the optimal solution of $L(G, w^t, I^t, k^t)$, thus

$$L(G^t, w^t, I^t, k^t) - L(G^{t+1}, w^t, I^t, k^t) \geq 0. \tag{41}$$

Similarly, I^{t+1} is the optimal solution of (21c), hence

$$L(G^{t+1}, w^{t+1}, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^t) \geq 0. \tag{42}$$

According to Theorem 2, (21b) and (21d) are strongly convex with respect to w and k. On the basis of Definition 5, we have

$$\begin{aligned} &L(G^{t+1}, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^t, k^t) \\ &\geq \frac{m}{2}\left\|w^t - w^{t+1}\right\|_2^2 + \nabla_w L|^T_{(G^{t+1}, w^{t+1}, I^t, k^t)}(w^t - w^{t+1}), \end{aligned} \tag{43}$$

and

$$\begin{aligned} &L(G^{t+1}, w^{t+1}, I^{t+1}, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1}) \\ &\geq \frac{n}{2}\left\|k^t - k^{t+1}\right\|_2^2 + \nabla_k L|^T_{(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1})}(k^t - k^{t+1}), \end{aligned} \tag{44}$$

where $m, n > 0$.

Since $(G^{t+1}, w^{t+1}, I^t, k^t)$ and $(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1})$ are the minimums with respect to (21b) and (21d),

$$\nabla_w L|^T_{(G^{t+1}, w^{t+1}, I^t, k^t)} = 0 \tag{45}$$

and

$$\nabla_k L|^T_{(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1})} = 0. \tag{46}$$

Then we have

$$L(G^{t+1}, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^t, k^t) \geq \frac{m}{2}\left\|w^t - w^{t+1}\right\|_2^2, \tag{47}$$

$$L(G^{t+1}, w^{t+1}, I^{t+1}, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1}) \geq \frac{n}{2}\left\|k^t - k^{t+1}\right\|_2^2. \tag{48}$$

Finally, we can deduce that

$$\begin{aligned} &L(G^t, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1}) \\ &\geq \frac{m}{2}\left\|w^t - w^{t+1}\right\|_2^2 + \frac{n}{2}\left\|k^t - k^{t+1}\right\|_2^2 \\ &\geq C(\left\|w^t - w^{t+1}\right\|_2^2 + \left\|k^t - k^{t+1}\right\|_2^2), \end{aligned} \tag{49}$$

where $C = min(\frac{m}{2}, \frac{n}{2})$. The proof is completed. □

Theorem 4 *Under the assumptions in Theorem 3, the sequences $\left\{L(G^t, w^t, I^t, k^t)\right\}$ and $\left\{G^t, w^t, I^t, k^t\right\}$ are bounded.*

Proof By the definitions $\|\mathcal{P}(I)\|_0$ and $\|G\|_0$, we know that $\|\mathcal{P}(I)\|_0 \geq 0$ and $\|G\|_0 \geq 0$. In addition, $\|k \otimes w - B\|_2^2$, $\|k\|_2^2$, $\|\nabla^\alpha I - G\|_2^2$, and $\|I - w\|_2^2$ are convex functions. So they have lower bounds. Now we can conclude that $L(G^t, w^t, I^t, k^t) > -\infty$. Based on the proof of Theorem 3, we have that $L(G^t, w^t, I^t, k^t) \leq L(G^0, w^0, I^0, k^0)$, which means $\left\{L(G^t, w^t, I^t, k^t)\right\}$ is bounded.

Next, we prove that $\left\{G^t, w^t, I^t, k^t\right\}$ is bounded. From Theorem 3, we have

$$\begin{aligned} &\left\|w^t - w^{t+1}\right\|_2^2 + \left\|k^t - k^{t+1}\right\|_2^2 \\ &\leq \frac{1}{C}(L(G^t, w^t, I^t, k^t) - L(G^{t+1}, w^{t+1}, I^{t+1}, k^{t+1})), \end{aligned} \tag{50}$$

and then

$$\begin{aligned} &\sum_{t=0}^{s}(\left\|w^t - w^{t+1}\right\|_2^2 + \left\|k^t - k^{t+1}\right\|_2^2) \\ &\leq \frac{1}{C}(L(G^0, w^0, I^0, k^0) - L(G^{s+1}, w^{s+1}, I^{s+1}, k^{s+1})). \end{aligned} \tag{51}$$

Due to $\left\{L(G^t, w^t, I^t, k^t)\right\}$ being bounded, when $s \to +\infty$, we can deduce that

$$\sum_{t=0}^{s}(\left\|w^t - w^{t+1}\right\|_2^2 + \left\|k^t - k^{t+1}\right\|_2^2) < +\infty, \tag{52}$$

hence $\sum_{t=0}^{\infty}\left\|w^t - w^{t+1}\right\|_2^2, \sum_{t=0}^{\infty}\left\|k^t - k^{t+1}\right\|_2^2 < +\infty$, w^t and k^t are bounded.

According to (45), we have

$${K^t}^T(K^t w^{t+1} - B) + \lambda_2(w^{t+1} - I^t) = 0, \tag{53}$$

which can imply

$$I^t = \frac{1}{\lambda_2}{K^t}^T(K^t w^{t+1} - B) + w^{t+1}. \tag{54}$$

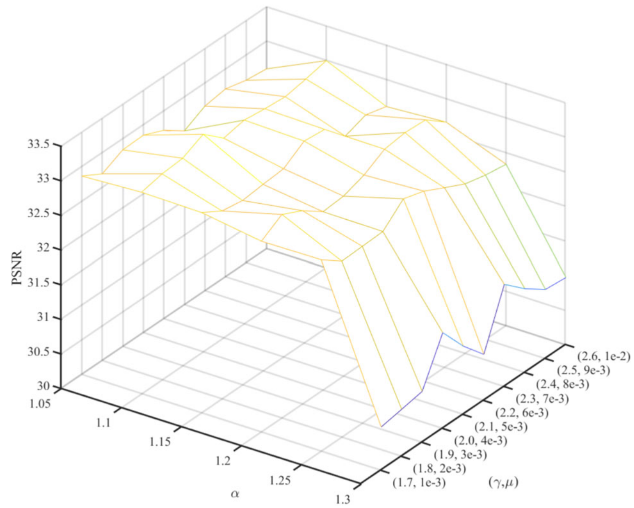


Fig. 3 PSNR values of the benchmark dataset [17] deconvolution by the proposed model with respect to parameters α, γ, and μ

Therefore, we can obtain

$$\begin{aligned} \|I^t\|_2^2 &= \frac{1}{\lambda_2}\|{K^t}^T(K^t w^{t+1} - B) + \lambda_2 w^{t+1}\|_2^2 \\ &\leq \frac{1}{\lambda_2}\|K^t\|_2^2\|K^t\|_2^2\left\|w^{t+1}\right\|_2^2 + \frac{1}{\lambda_2}\|K^t\|_2^2\|B\|_2^2 + \left\|w^{t+1}\right\|_2^2, \end{aligned} \tag{55}$$

due to the boundedness of k^t and w^t, and B being a constant matrix, thus $\left\|I^t\right\|_2^2 < +\infty$ and I^t is bounded.

Combining with

$$G^{t+1} = \begin{cases} 0, & (\nabla^\alpha I^t)^2 < \mu/\lambda_1, \\ \nabla^\alpha I^t, & otherwise, \end{cases} \tag{56}$$

we have

$$\begin{aligned} \left\|G^{t+1}\right\|_2^2 &\leq \|\nabla^\alpha I^t\|_2^2 = \left\|\begin{pmatrix} M_\alpha I^t \\ I^t {N_\alpha}^T \end{pmatrix}\right\|_2^2 \\ &= \|M_\alpha I^t\|_2^2 + \left\|I^t {N_\alpha}^T\right\|_2^2 \\ &\leq \|M_\alpha\|_2^2\|I^t\|_2^2 + \|I^t\|_2^2\|N_\alpha\|_2^2, \end{aligned} \tag{57}$$

due to the boundedness of I^t, M_α, and N_α being constant matrices, thus $\left\|G^{t+1}\right\|_2^2 < +\infty$, G^t is bounded.

So we can deduce the conclusion, $\left\{L(G^t, w^t, I^t, k^t)\right\}$ and $\left\{G^t, w^t, I^t, k^t\right\}$ are bounded. □

Theorem 5 *$\left\{G^t, w^t, I^t, k^t\right\}$ has at least a limit point $\{G^*, w^*, I^*, k^*\}$ and $\left\{L(G^t, w^t, I^t, k^t)\right\}$ converges to $L(G^*, w^*, I^*, k^*)$ based on Algorithm.*

Proof From Theorem 4, the sequence $\left\{G^t, w^t, I^t, k^t\right\}$ is bounded. Then there exist a convergent subsequence and

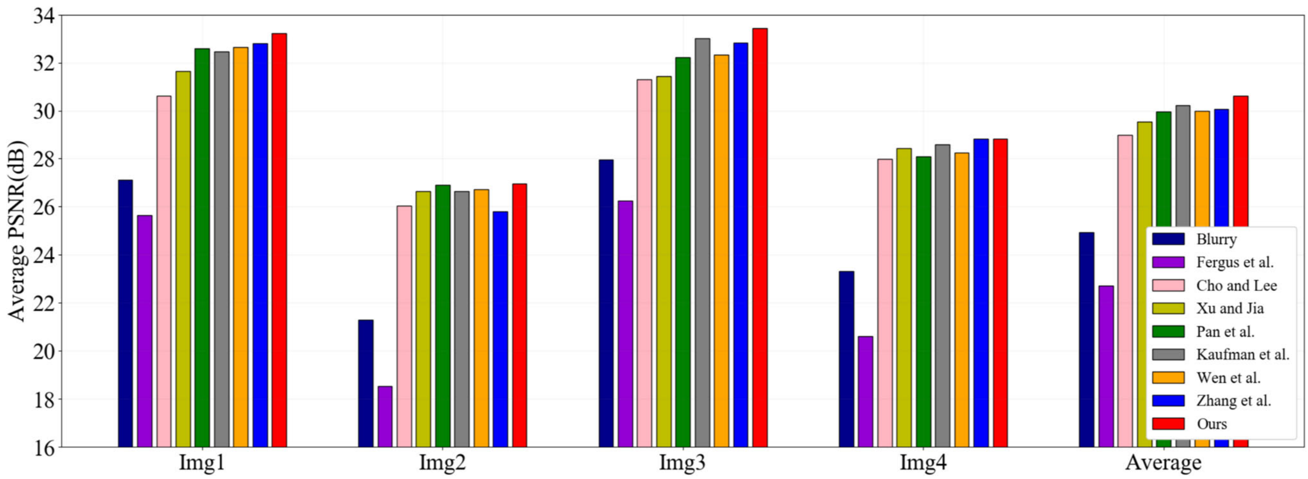


Fig. 4 Quantitative evaluation results on the benchmark dataset [8]. We calculated the average PSNR of each image under 12 blur kernels for different methods

Table 1 Quantitative results on the köhler dataset [8]

Method	PSNR(dB)	SSIM
Fergus et al. [16]	22.73	0.7048
Cho and Lee [1]	28.98	0.8746
Xu and Jia [4]	29.54	0.8851
Krishnan et al. [44]	25.72	0.7608
Pan et al. [3]	29.95	0.8853
Kaufman et al. [26]	<u>30.22</u>	0.8935
Wen et al. [19]	29.98	<u>0.8944</u>
Zhang et al. [42]	30.06	0.8867
Ours	**30.62**	**0.8951**

Bold text is used to denote the optimal outcomes, while underlined text indicates the next best results

Table 2 Assessing fractional-order ($\boldsymbol{F}$)'s contribution through ablation analysis in DCP and PMP Deconvolution

Method	DCP		PMP	
	w/o $\boldsymbol{F}$	w/ $\boldsymbol{F}$	w/o $\boldsymbol{F}$	w/ $\boldsymbol{F}$
PSNR(dB)	29.95	30.35	29.98	30.62
SSIM	0.8853	0.8924	0.8944	0.8951

a limit point denoted by $\{G^*, w^*, I^*, k^*\}$. Then by Theorem 3 and Theorem 4, $\{L(G^t, w^t, I^t, k^t)\}$ is a monotonically decreasing sequence and lower bounded. According to the monotone convergence theorem, we can state that $L(G^t, w^t, I^t, k^t) \rightarrow L(G^*, w^*, I^*, k^*)$ as $t \rightarrow +\infty$. $\square$

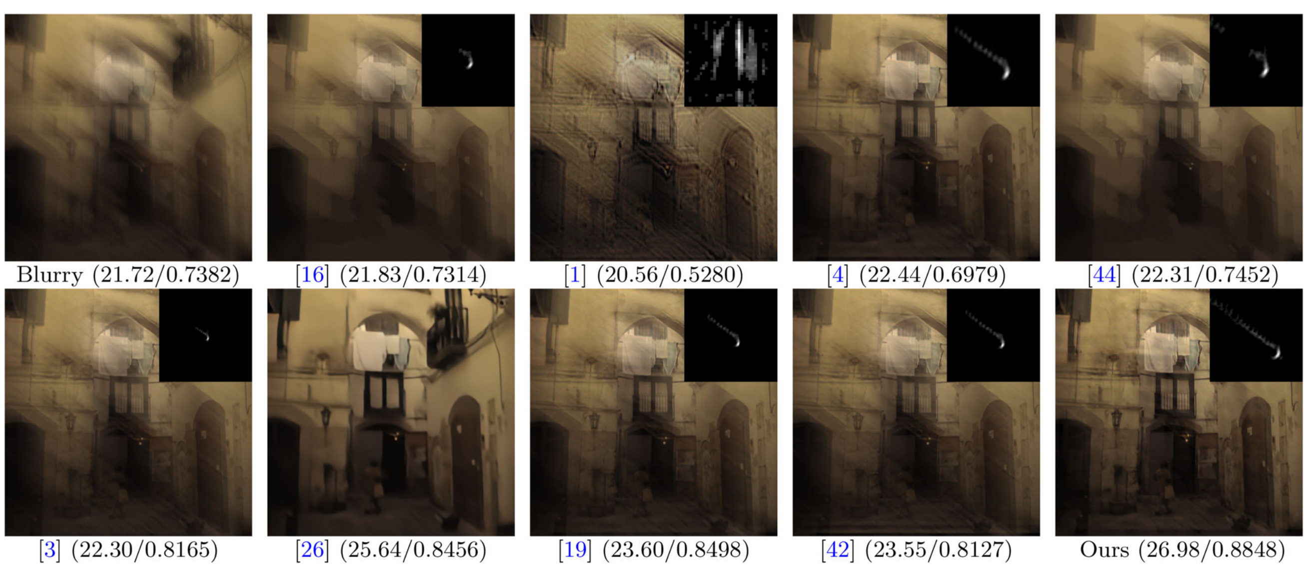


Fig. 5 Visual comparisons with the advanced methods on the dataset [8]. The blur kernel is in the top right corner of each image, and the PSNR/SSIM value is at the bottom of each image. When compared to the latest techniques, our approach excels in achieving the clearest results

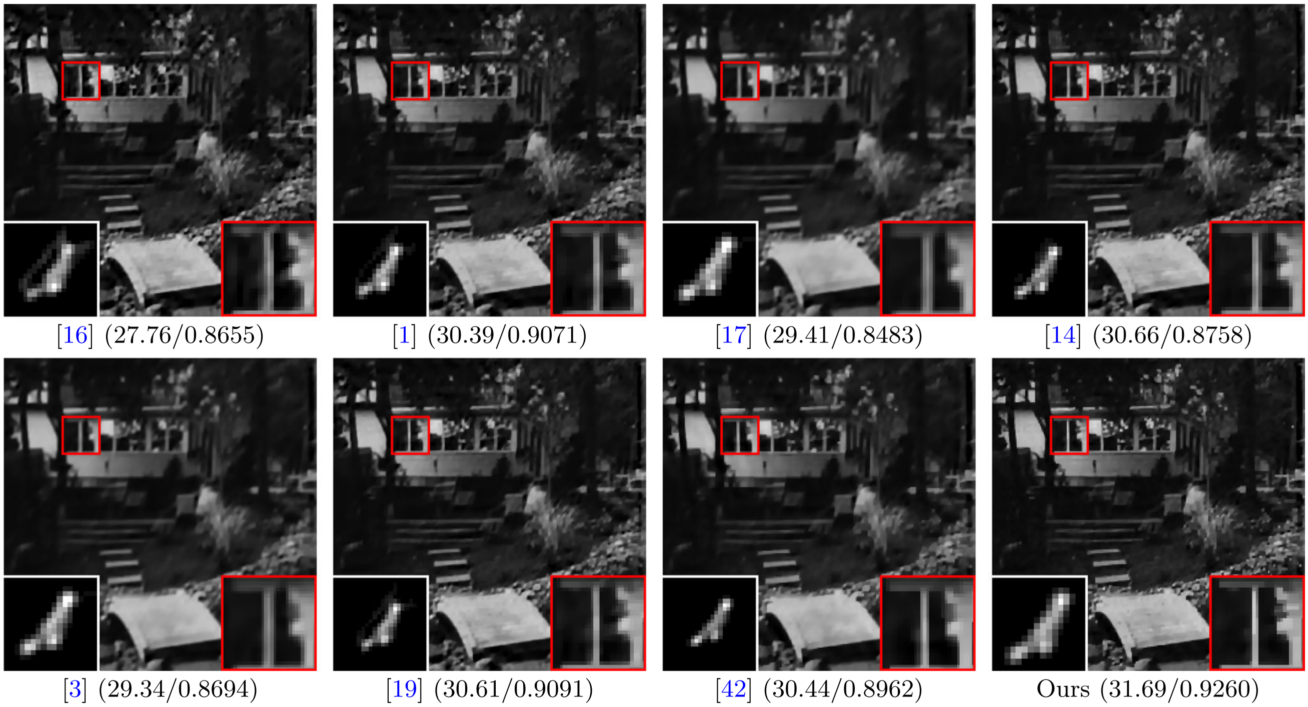


Fig. 6 Deconvolution visual results of different methods on the dataset [17], along with the corresponding PSNR/SSIM score. The white box is the blur kernel, and the red box is the enlarged region. Our approach demonstrates its superior effectiveness in comparison to the state-of-the-art methods, yielding the sharpest image results

6 Experimental Results

In this section, we show the experimental results of the proposed model, including objective evaluation and the visual effects compared with other methods. We commence by evaluating our method on Köhler et al. [8], Levin et al. [17] and Lai et al. [10] datasets to demonstrate its efficacy. Subsequently, we explore its robustness and applicability using real-world blurred images.

Köhler et al dataset Generated through precise camera motion tracking, it provides authentic and reproducible blurring conditions, suitable for evaluating the performance of deconvolution algorithms on specific types of motional blur.

Levin et al dataset It is widely recognized in image deconvolution for its clear and synthetically blurred images, enabling controlled parameter manipulation and algorithm performance evaluation.

Lai et al dataset It includes a diverse range of real-world blurred images paired with their corresponding clear images, which is appropriate for assessing the performance of deconvolution algorithms on complex, real-world blurry images.

Real-World Images dataset It contains only blurred images without clear references, which are more reflective of real-world application scenarios. This dataset poses a higher level of challenge and is suitable for evaluating the robustness of deconvolution algorithms in handling unknown blurring conditions.

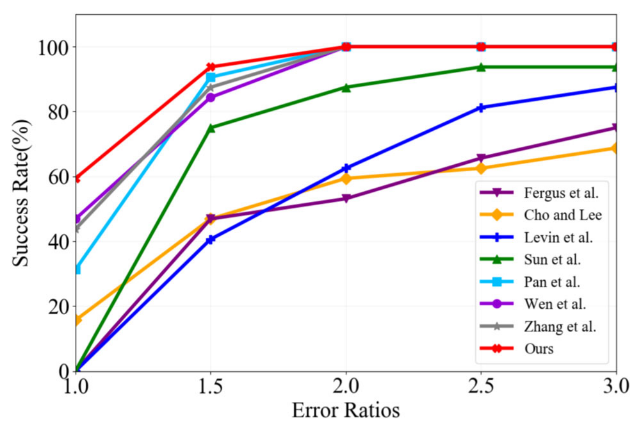
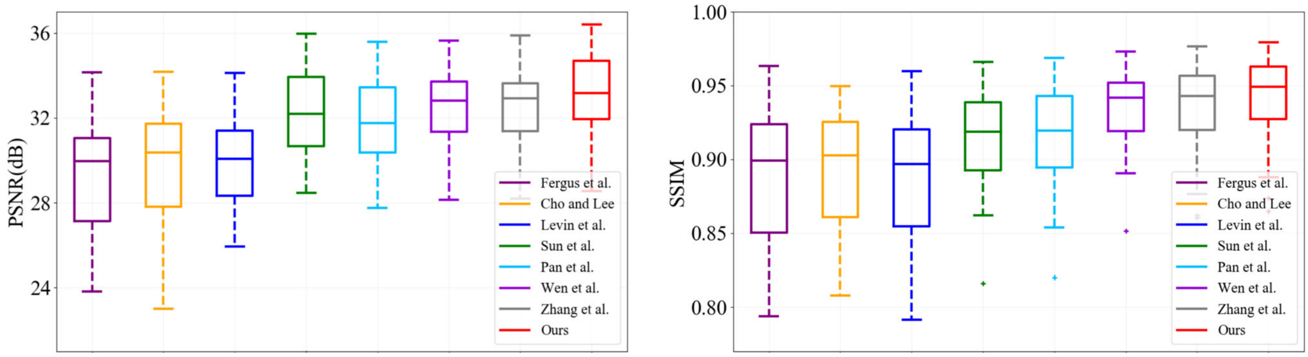


Fig. 7 Error ratios on dataset [17]. The method demonstrates superior performance when a higher success rate is achieved, assuming a fixed error rate

To objectively assess the performance of the proposed model in deconvolution, we utilize four metrics: the peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM) [45], the sum of squared differences (SSD), and Multi-dimension Attention Network for no-reference Image Quality Assessment (MANIQA) [46]. These metrics serve as quantitative measures to evaluate the deconvolution capabili-



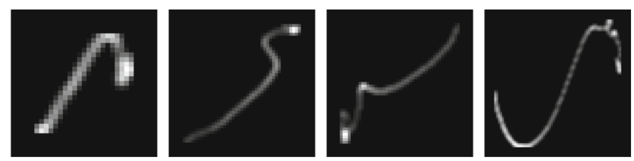
(a) PSNR (b) SSIM

Fig. 8 Comparison of averaged PSNR and SSIM on dataset [17]. The box plots demonstrate that our method consistently outperforms the other methods in terms of both PSNR and SSIM metrics

Table 3 Quantitative results on the Levin dataset [17]

Method	PSNR(dB)	SSIM
Fergus et al. [16]	29.46	0.8451
Cho and Lee [1]	30.79	0.8837
Levin et al. [17]	31.14	0.8960
Sun et al. [14]	32.10	0.9033
Pan et al. [3]	31.73	0.9148
Wen et al. [19]	32.45	0.9344
Zhang et al. [42]	32.57	0.9362
Ours	**33.14**	**0.9405**

Bold text is used to denote the optimal outcomes, while underlined text indicates the next best results



(a) kernel01 (b) kernel02 (c) kernel03 (d) kernel04

Fig. 9 Blur kernels applied in the creation of the 100 artificially generated uniformly blurred images as detailed in Lai et al. [10]. The shape of the blur kernels varies significantly in terms of thickness and length

ties of the model, and a higher value of these metrics indicates better image quality. The initial three metrics constitute full-reference image quality assessment indices. In contrast, the latter belongs to a specialized class of no-reference image quality assessment metrics designed for evaluating images captured in real-world scenarios without ground truth.

All experiments were executed on a personal computer with an Intel Core i5 CPU 12490F and 12 GB of RAM, using MATLAB R2021a (Windows 10) as the development environment. For comparative assessment, we compared our method against several state-of-the-art methods, including recent deep learning methods.

For our experiments, parameter settings were defined as: $\mu = \lambda_1 = \lambda_2 = 4 \times 10^{-3}$, $\gamma = 2$, and $max\_iter = 5$. Considering the convergence rate and computational time, we set the growth multiple η as 2. Besides, the fractional-order α remained constant at 1.1. It is imperative to underscore that optimal parameter values may be influenced by various factors, including image size, saturation, and the specific optimization algorithm employed. In blind deconvolution, parameter selection is often an empirical process influenced by experiential learning and iterative testing. In Fig. 3, we also discuss the influence of fractional-order α, positive weight parameters γ and μ.

6.1 Köhler et al. Dataset

Utilizing the Köhler dataset [8], which comprises 4 images and 12 corresponding blur kernels, we compared our approach with notable methods such as [1, 3, 4, 16, 19, 26, 42, 44]. The evaluation metrics employed were the PSNR and SSIM, with higher values of both indicating superior deconvolution.

Figure 4 illustrates that our method consistently outperforms other methods in terms of PSNR across the entire Köhler dataset. Moreover, the average SSIM values presented in Table 1 further validate the superior performance of our method. These results demonstrate that our approach produces the highest quality deconvolution images in terms of both PSNR and SSIM.

Figure 5 visually compares four challenging images with pronounced blur, sourced from the dataset [8]. These images, namely 'Blurry3_9,' are subjected to our proposed algorithm as well as sophisticated methods. By effectively reducing blur artifacts and enhancing image details, our method produces visually pleasing results that are on par with or surpass the performance of existing methods.

Table 2 delves deeper into assessing the efficacy of the proposed fractional-order variational model, presenting outcomes for both DCP with and without fractional order, as well as PMP with and without fractional order on the benchmark dataset [8]. It is evident that the absence of fractional orders, i.e., integer orders, exhibits significantly inferior performance compared to fractional orders.

6.2 Levin et al. Dataset

We additionally assess our approach using the dataset provided by Levin et al. [17], comprising four images and eight kernels. We compare our proposed strategy against cutting-edge algorithms, including those presented in references [1, 3, 14, 16, 17, 19, 42]. Firstly, we demonstrate the robustness of our algorithm through the SSD, which measures the sum of the squared differences between corresponding pixel intensities of the clean and blurred images. A smaller SSD value indicates a higher similarity between the images, while a larger SSD value indicates greater dissimilarity. As shown in Fig. 7, almost 60% of our results have calculated SSD error ratio below 1.0, and the result with an SSD error ratio below 2.0 reaches 100%. Moreover, Table 3 illustrates the mean PSNR and SSIM values for different methods, and Fig. 8 exhibits a statistical evaluation of the PSNR and SSIM outcomes.

From Table 3, it can be seen that our results have the highest PSNR and SSIM values; in other words, our algorithm is robust. Additionally, as shown in Fig. 8, we have the highest median and relatively concentrated data points. We present a visual performance evaluation in Fig. 6. Our method outperforms others by alleviating the ringing artifacts and achieving the highest PSNR.

6.3 Lai et al. Dataset

To provide additional evidence of our method's viability, we conducted a separate set of experiments using the more demanding datasets recommended by Lai et al. [10]. In this section, we assess the performance of our proposed approach on a collection of 100 blurred images. Within this dataset, each of the four blur kernels as shown in Fig. 9 is paired with 25 reference images, categorized into five distinct groups: manmade, natural, people, saturated, and text. In conjunction with the dataset, we achieve comparable or even better visual results compared with existing state-of-the-art methods.

In Tables 4-5, we delve deeper into the quantitative analysis of outcomes produced through different approaches. The average PSNR and SSIM between the deconvolution images and the original clean images were utilized as benchmarks for evaluation. As evidenced by the table, our approach predominates in terms of performance across most scenarios. We present a visual performance comparison in Fig. 10.

6.4 Real-World Images

In addition, we evaluated our method more using images captured from real-world scenes such as faces, natural scenes, text, and low-light conditions. In the absence of a clear reference image, we utilize the MANIQA as an objective metric for image quality assessment.

Face image Fig. 11 showcases a visual comparison of various deconvolution methods applied to the face images. The top-performing approach made by Wen et al. [19] strikes a pleasing balance between edge sharpness and overall smoothness, but there is still noticeable local blur. In contrast, our algorithm leverages the combined power of PMP and fractional-order derivatives, effectively retaining the strengths of PMP while augmenting its performance. As shown, we are capable of reconstructing facial features that closely resemble the actual features. Specifically, our approach excels at retaining the integrity of facial features—such as the eyes, nose, and mouth—while significantly mitigating the blurring effects typically introduced by motion.

Natural image To evaluate the significance of our method on natural blurred images, we conducted tests and comparisons. Figure 12 demonstrates our utilization of the same non-blind deconvolution method to produce the outcomes for each compared method [3, 19, 42, 52]. The other comparison methods have more or less amplified the ringing artifacts. In contrast, our method enhances edges and textures through algorithmic refinement while maintaining the smoothness of the background, thereby increasing clarity without compromising the natural feel of the image.

Text image In the study presented in [19], the inclusion of the PMP prior has proven advantageous for blind deconvolution in text images. To validate the effectiveness of our proposed method, we conduct additional experiments specifically focusing on text images. Notably, text images possess distinctive attributes such as sharp edges, high contrast, and often a relatively simple background, which present unique challenges for deconvolution algorithms. Our method leverages fractional-order derivatives, which are particularly adept at capturing the edge details in text images. Figure 13 showcases the remarkable performance of our approach when compared to [3, 19, 42, 52]. As shown, the text information we retrieve is highly recognizable and fluent.

Low-light image In Fig. 14, we showcase a sample low-illumination image extracted from Hu et al.'s dataset [13]. Conventional natural image deconvolution techniques struggle to produce clean results due to the influence of large saturated areas. Due to the nature of integer-order properties, the red-bordered regions in other methods exhibit false edges around abrupt transitions. In contrast, our approach delivers a comparable outcome to other methods. As shown, our

Table 4 Comparisons of averaged PSNR on Lai dataset [10]

PSNR(dB)	kernel01	kernel02	kernel03	kernel04
Fergus et al. [16]	16.66	15.67	15.25	14.31
Cho and Lee [1]	19.07	17.65	18.06	15.19
Xu and Jia [4]	22.64	21.65	20.95	17.72
Levin et al. [47]	19.45	18.77	18.95	16.78
Zhang et al. [48]	21.25	18.58	19.21	16.59
Xu et al. [18]	22.11	20.96	19.96	16.64
Michaeli et al. [49]	21.44	19.33	18.90	16.00
Perrone and Favaro [50]	21.28	18.13	19.15	18.27
Pan et al. [3]	23.64	21.44	21.50	18.46
SelfDeblur [51]	21.38	20.54	20.24	18.50
Chen et al. [12]	23.80	21.96	21.90	19.60
Kaufman et al. [26]	24.24	22.06	21.64	19.82
Wen et al. [19]	23.05	20.73	21.16	18.13
Zhang et al. [42]	<u>24.43</u>	<u>23.45</u>	<u>22.76</u>	<u>21.22</u>
Ours	**25.86**	**23.98**	**24.22**	**21.86**

Bold text is used to denote the optimal outcomes, while underlined text indicates the next best results

Table 5 Comparisons of averaged SSIM on Lai dataset [10]

SSIM	kernel01	kernel02	kernel03	kernel04
Fergus et al. [16]	0.5429	0.4604	0.4571	0.3346
Cho and Lee [1]	0.6846	0.5892	0.6251	0.4383
Xu and Jia [4]	0.8346	0.8096	0.7739	0.6128
Levin et al. [47]	0.7287	0.6812	0.6923	0.5614
Zhang et al. [48]	0.7826	0.6172	0.6802	0.4994
Xu et al. [18]	0.8102	0.7765	0.7185	0.5736
Michaeli et al. [49]	0.7774	0.6547	0.6470	0.4473
Perrone and Favaro [50]	0.8191	0.6460	0.7295	0.6474
Pan et al. [3]	0.8776	0.8092	0.8128	0.6385
SelfDeblur [51]	0.7840	0.7550	0.7360	0.5890
Chen et al. [12]	0.8827	0.8242	0.8349	0.7120
Kaufman et al. [26]	<u>0.8951</u>	0.8045	0.7913	0.6629
Wen et al. [19]	0.8608	0.7634	0.7995	0.6391
Zhang et al. [42]	0.8941	<u>0.8522</u>	<u>0.8625</u>	<u>0.7817</u>
Ours	**0.9181**	**0.9102**	**0.8805**	**0.8108**

Bold text is used to denote the optimal outcomes, while underlined text indicates the next best results

method remains unaffected in low-light conditions, enabling us to recover clean images without any artifacts .

6.5 Computational Complexity

In contrast with $\mathcal{L}_0$ regularized methods [18], our approach additionally requires the computation of the fractional order and PMP. For the fractional order, the computational complexity primarily arises from matrix multiplications, resulting in a complexity of $O\left(m^2n + mn^2\right)$. For the PMP, the initial step involves partitioning the image into P smaller blocks, each with dimensions $r \times r$. The total number of blocks P is obtained through the expression $P = \lceil \frac{m}{r} \rceil \times \lceil \frac{n}{r} \rceil$. The complexity of processing each block is $O\left(r^2\right)$. We can obtain the complexity of the PMP is $O\left(P \times r^2\right)$, which simplifies to $O\left(mn\right)$. Thus the overall order of complexity per iteration of Algorithm 1 is $O\left(m^2n + mn^2\right)$.

6.6 Limitations

While the blind deconvolution method based on the fractional-order variational model proposed in this study has achieved significant success in handling uniformly blurred images, it must be acknowledged that the method's scope is limited to scenarios of uniform blurring. This implies that for more complex spatially variant blurs, such as defocus blur, the

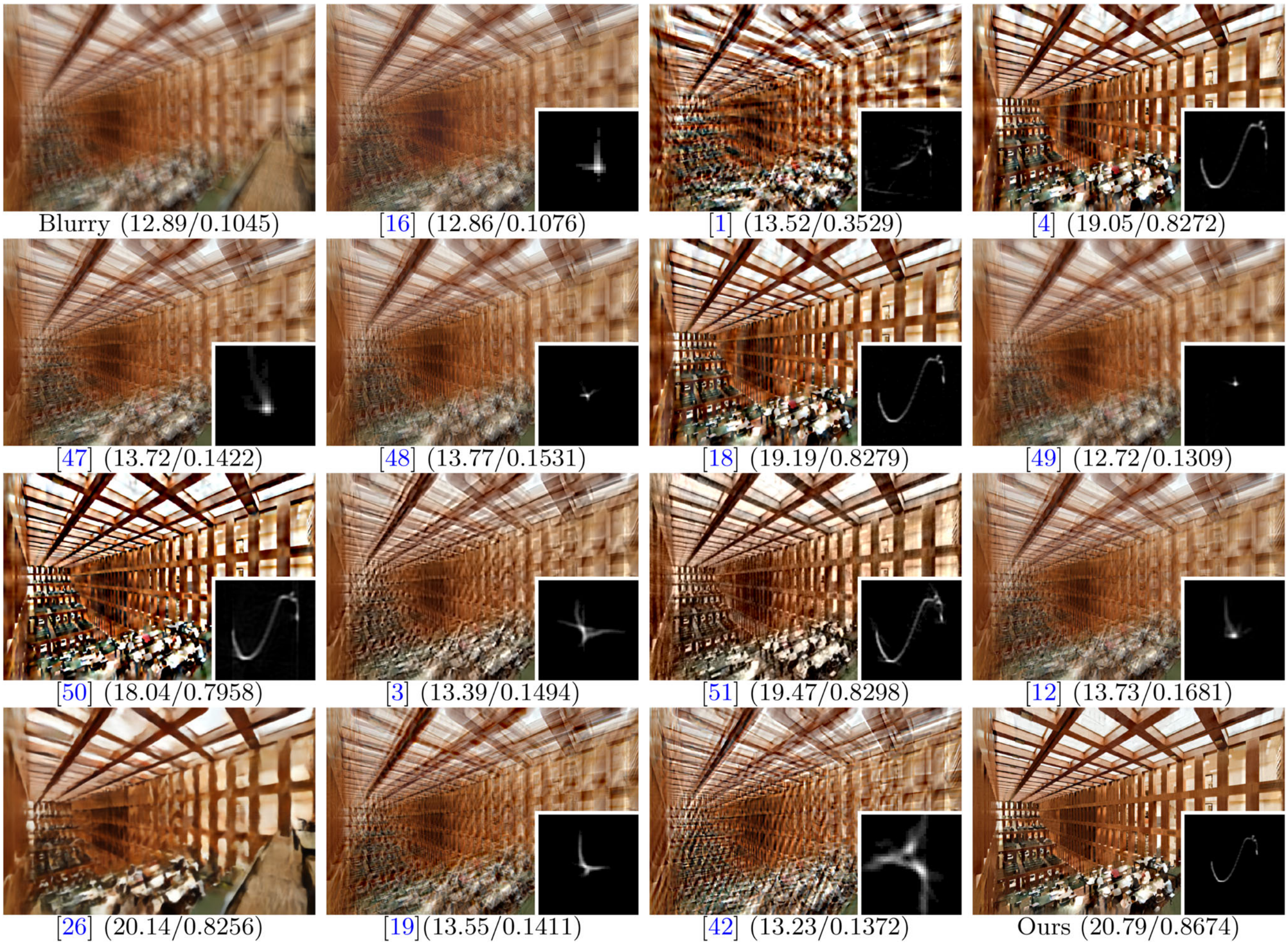


Fig. 10 Visual examples produced by notable blind deconvolution methods and our proposed approach, along with the corresponding PSNR/SSIM score

method may not be directly applicable or achieve the high level of deconvolution effectiveness. Therefore, future work will focus on extending this method to adapt to a broader range of blur types.

7 Conclusions

In this study, we introduce a novel method for addressing blurred images by proposing a fractional-order variational model based on the PMP prior. This fractional-order variational model is shown to preserve more intricate image information while reducing the ringing artifacts observed in deconvolution images.

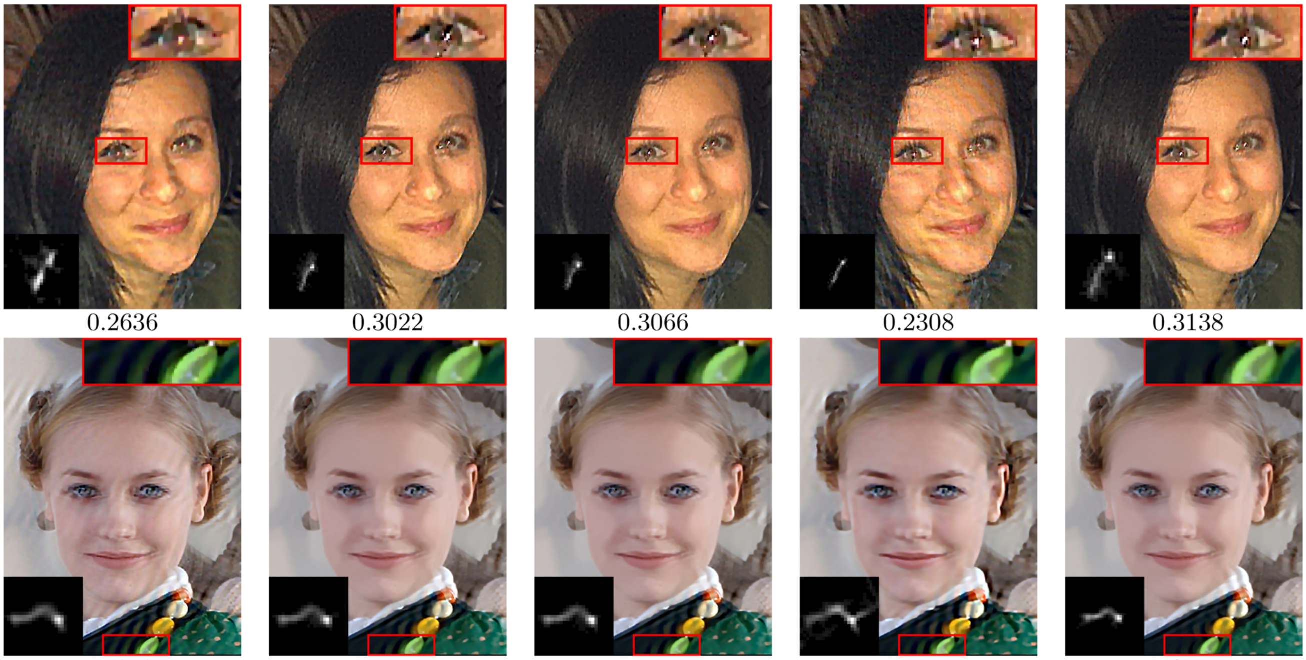
To further augment the regularization process of fractional-order images, we design an efficient optimization algorithm based on the half-quadratic strategy. This algorithm optimizes the process of computing image gradients within the variational model, bolstering the deconvolution procedure by promoting the production of sharp images over blurred ones. Then, we prove the convergence of the algorithm and the monotonic decrease in the objective function values.

Our results suggest convincing evidence of the effectiveness of our method. Our approach not only performs favorably compared to other blind deconvolution methods, but also shows the potential to push the boundaries of current image deconvolution practices. Therefore, this study presents a significant stride toward resolving real-world deconvolution challenges, which have implications for various applications where image clarity is paramount.

Appendix A Proof of (22)

Proof For the problem (21a), we can split it into element-wise forms

$$\sum_{i}^{m}\sum_{j}^{n}\min\,(\nabla^{\alpha} I_{i,j}{}^{t} - G_{i,j})^2 + \frac{\mu}{\lambda_1} H(G_{i,j}), \qquad \text{(A.1)}$$

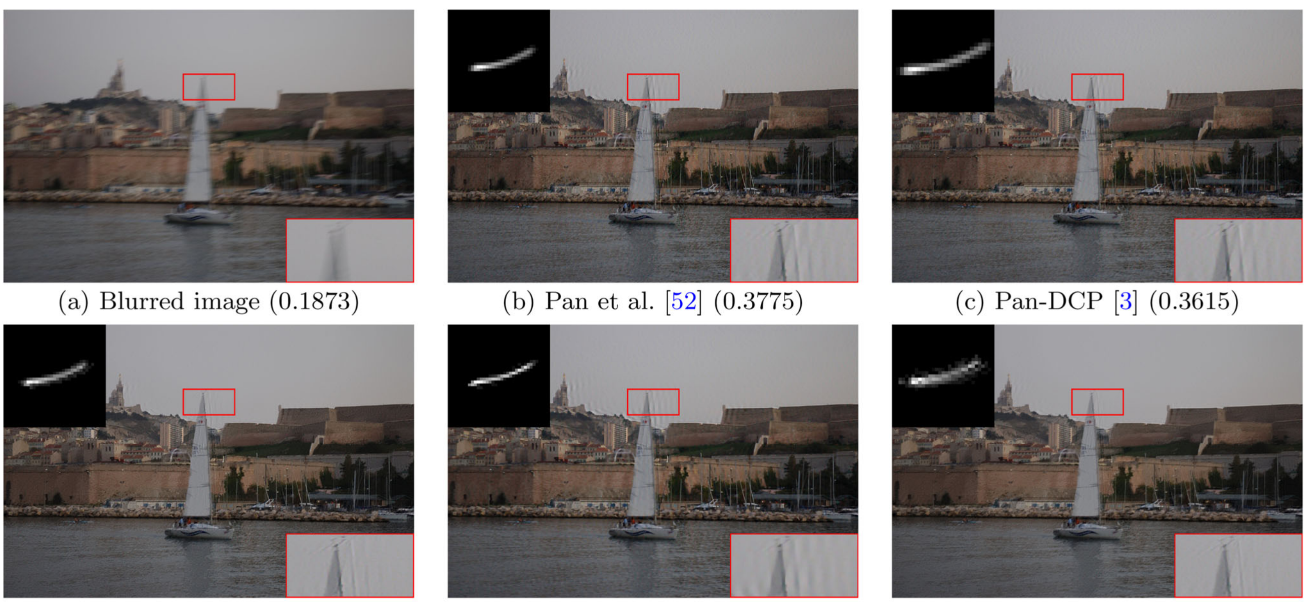


0.2636 0.3022 0.3066 0.2308 0.3138

0.3717 0.3966 0.3958 0.3822 0.4932

(a) Pan et al. [52] (b) Pan-DCP [3] (c) Wen et al. [19] (d) Zhang et al. [42] (e) Ours

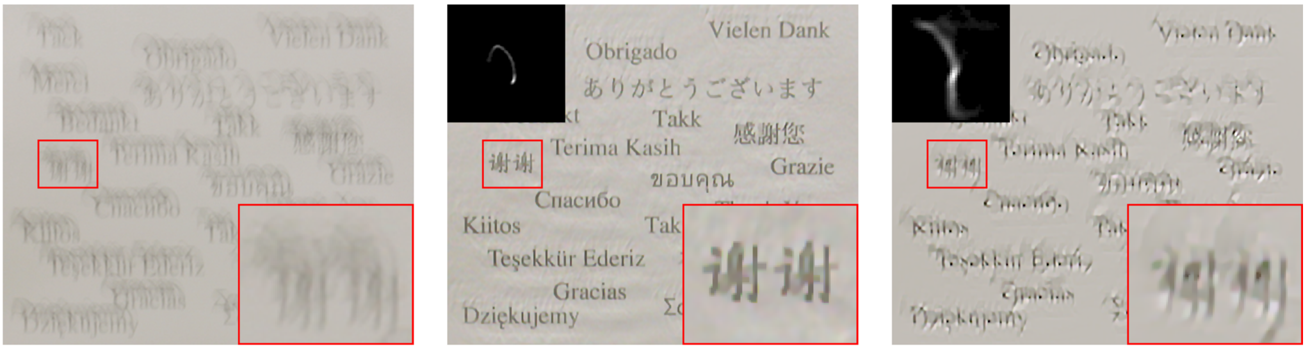
Fig. 11 Comparisons on two realistic blurred face images, along with the corresponding MANIQA (↑) score. Our method generates visually comparable or even better deconvolution results than the reported results of [3, 19, 42, 52]



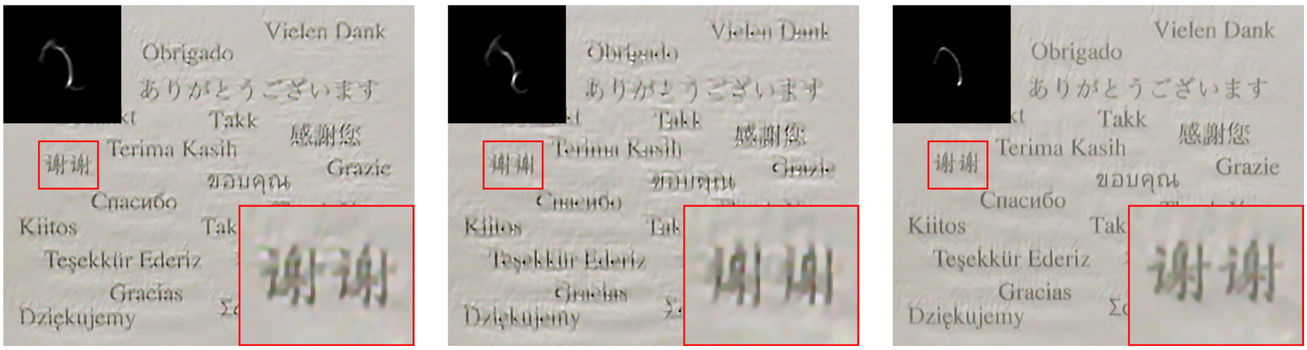
(a) Blurred image (0.1873) (b) Pan et al. [52] (0.3775) (c) Pan-DCP [3] (0.3615)

(d) Wen et al. [19] (0.4543) (e) Zhang et al. [42] (0.2992) (f) Ours (0.5046)

Fig. 12 Comparisons on nature image, along with the corresponding MANIQA (↑) score. Our method generates visually comparable or even better deconvolution results than the reported results of [3, 19, 42, 52]

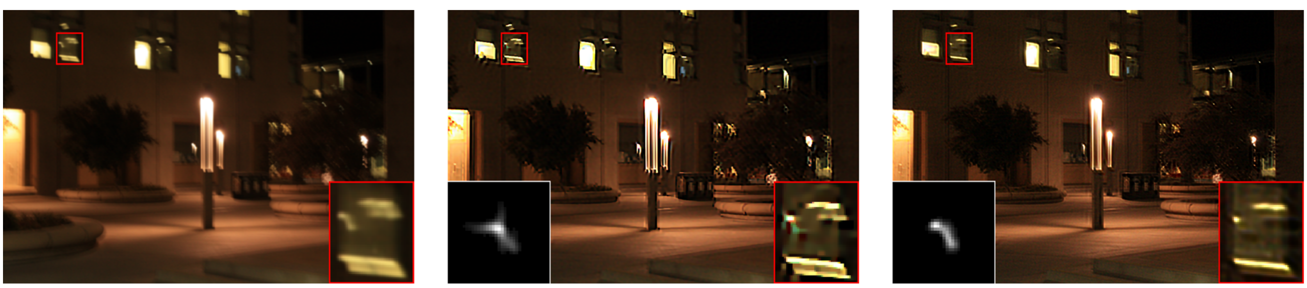


(a) Blurred image (0.2015) (b) Pan et al. [52] (0.2829) (c) Pan-DCP [3] (0.1345)

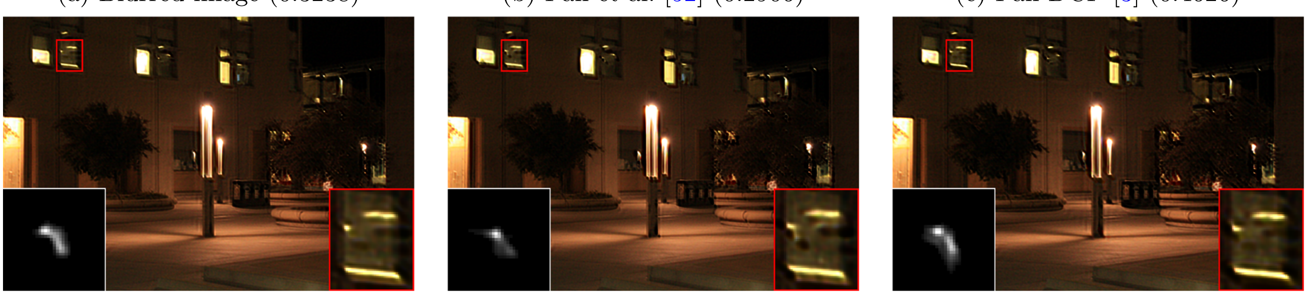


(d) Wen et al. [19] (0.2296) (e) Zhang et al. [42] (0.1936) (f) Ours (0.3263)

Fig. 13 Comparisons on the text image, along with the corresponding MANIQA (↑) score. Our method generates visually comparable or even better deconvolution results than the reported results of [3, 19, 42, 52]



(a) Blurred image (0.3238) (b) Pan et al. [52] (0.2906) (c) Pan-DCP [3] (0.4626)



(d) Wen et al. [19] (0.4586) (e) Zhang et al. [42] (0.4400) (f) Ours (0.4819)

Fig. 14 Comparisons on low-light image deconvolution, along with the corresponding MANIQA (↑) score. Our method generates visually comparable or even better deconvolution results than the reported results of [3, 19, 42, 52] where $H(G_{i,j})$ is a binary function returning 1 if $G_{i,j} \neq 0$ and 0 otherwise. For a single element, we can establish the equation

$$\mathbb{S}_{i,j} = (\nabla^\alpha I_{i,j}{}^t - G_{i,j})^2 + \frac{\mu}{\lambda_1} H(G_{i,j}). \tag{A.2}$$

When $(\nabla^\alpha I_{i,j}{}^t)^2 < \mu/\lambda_1$ and $G_{i,j} = 0$, we have $\mathbb{S}_{i,j} = (\nabla^\alpha I_{i,j}{}^t)^2$. Note that $G_{i,j} \neq 0$, then

$$\mathbb{S}_{i,j} = (\nabla^\alpha I_{i,j}{}^t - G_{i,j})^2 + \frac{\mu}{\lambda_1} \geq \frac{\mu}{\lambda_1}. \tag{A.3}$$

Therefore, we have $\min \mathbb{S}_{i,j} = (\nabla^\alpha I^t_{i,j})^2$ when $G_{i,j} = 0$.

When $(\nabla^\alpha I_{i,j}{}^t)^2 \geq \mu/\lambda_1$ and $G_{i,j} = 0$, $\mathbb{S}_{i,j} = (\nabla^\alpha I_{i,j}{}^t)^2$. Note that $G_{i,j} \neq 0$, then

$$\mathbb{S}_{i,j} = (\nabla^\alpha I_{i,j}{}^t - G_{i,j})^2 + \frac{\mu}{\lambda_1} = \frac{\mu}{\lambda_1} \tag{A.4}$$

as $G_{i,j} = \nabla^\alpha I_{i,j}{}^t$. Thus $\min \mathbb{S}_{i,j} = \frac{\mu}{\lambda_1}$ when $G_{i,j} = \nabla^\alpha I_{i,j}{}^t$.

As a consequence, when

$$G_{i,j} = \begin{cases} 0, & (\nabla^\alpha I^t_{i,j})^2 < \mu/\lambda_1, \\ \nabla^\alpha I^t_{i,j}, & otherwise, \end{cases} \tag{A.5}$$

$\mathbb{S}_{i,j}$ has a minimum value, and then for the problem (21a), the solution can be explicitly obtained as follows:

$$G^{t+1} = \begin{cases} 0, & (\nabla^\alpha I^t)^2 < \mu/\lambda_1, \\ \nabla^\alpha I^t, & otherwise. \end{cases} \tag{A.6}$$

□

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Data Availability Enquiries about data availability should be directed to the authors.

Declarations

Conflict of interest The authors have no conflict of interest to declare that is relevant to the content of this article.

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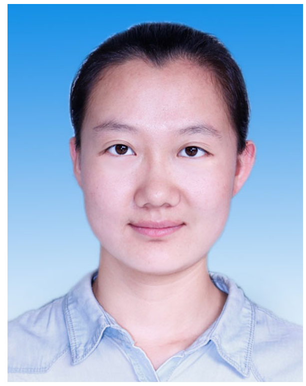

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